BOARD OF INTERMEDIATE EDUCATION, KARACHI

INTERMEDIATE (12TH CLASS) EXAMINATION, 2021

PHYSICS

Past Paper 2021

Retyped faithfully from the original board paper — same questions, no changes.

Section 'A' (Multiple Choice Questions) (42)

Maximum Marks: 42

Note: This section consists of 42 part questions and all are to be answered. Each question carries ONE mark. Do not copy the part questions in your answer book. Write only the answer in full against the proper number of the question and its part. (Options in the original are separated by star marks, shown here as A–D in printed order.)

Note: Time: 2 Hours — Maximum Marks: 85 (As per condensed syllabus)

Q1. Select the correct answer for each from the given options:

Qi. The mass of fissionable material needed for self-sustaining chain reaction is called:

(A) Atomic mass
(B) Critical mass
(C) Sub-critical mass
(D) Super critical mass

Qii. Transistor is a semi conducting device which has :

(A) Two terminals
(B) Three terminals
(C) Four terminals
(D) Five terminals

Qiii. In order to increase the number of electrons in photoelectric effect, this should be increased:

(A) Intensity of light
(B) Threshold frequency
(C) Work function
(D) Wavelength of light

Qiv. The magnitude of drift speed of electron is of the order of:

(A) 0.1 m/s
(B) 0.01 m/s
(C) 0.001m/s
(D) 0.0001m/s

Qv. For an ideal gas at constant temperature, the graph between V and 1/p is:

(A) Hyperbola
(B) Parabola
(C) Straight line
(D) Circle

Qvi. The life time of metastable state is of the order of:

(A) 10^{-8}s
(B) 10^{-4}s
(C) 10^{-3}s
(D) 10^{-2}s

Qvii. X – rays are produced if heavier atoms are bombarded by:

(A) Protons
(B) Electrons
(C) Neutrons
(D) Positrons

Qviii. This quantity is called Compton's wavelength:

(A) h/m_oc^2
(B) hc/m_o
(C) h/m_oc
(D) m_oc/h

Qix. This factor is Lorentz factor:

(A) $1-v^2/c^2$
(B) $\sqrt{(1-v^2/c^2)}$
(C) $\sqrt{(1-c^2/v^2)}$
(D) $\sqrt{(v^2/c^2-1)}$

Qx. Transistor in a circuit is used as:

(A) An oscillator
(B) An amplifier
(C) A half wave rectifier
(D) A full wave rectifier

Qxi. To convert galvanometer into voltmeter:

(A) A high value resistance is connected in series
(B) A high value resistance is connected in parallel
(C) A low value resistance is connected in series
(D) A low value resistance is connected in parallel

Qxii. The sensitivity of a galvanometer can be increased by:

(A) Increasing magnetic field
(B) Decreasing area of coil
(C) decreasing of number of turns in the coil
(D) making suspension wire thick

Qxiii. In step down transformer:

(A) $N_s > N_p$
(B) $N_s < N_p$
(C) $N_s = N_p$
(D) $N_s/N_p = 1$

Qxiv. Electrical conductor contains:

(A) only free electrons
(B) only bounded electrons
(C) both free bounded electrons
(D) Neither free nor bounded electrons

Qxv. The capacitance of a parallel plate capacitor does not depend upon:

(A) Area of plates
(B) Nature of plates
(C) Distance between the plates
(D) Medium between the plates

Qxvi. The electric field intensity between two similarity charge parallel plates is:

(A) σ/ε_0
(B) $\sigma/2\varepsilon_0$
(C) Zero
(D) $2\sigma/\varepsilon_0$

Qxvii. In amplitude modulation, amplitude of a wave:

(A) only increase
(B) only decrease
(C) increases as well as decreases
(D) neither increases nor decreases

Qxviii. The quantity of heat required to raise the temperature of unit mass of substance through one degree centigrade is called:

(A) Molar specific heat
(B) Heat capacity
(C) Specific heat
(D) Latent heat

Qxix. A thermodynamic process in which heat exchange is not allowed is called:

(A) Isochoric
(B) Isobaric
(C) Isothermal
(D) Adiabatic

Qxx. The atomic number of a radioactive element is increased by the emission of:

(A) $\alpha -$ particle
(B) negative β particle
(C) positive β particle
(D) $\gamma -$ rays

Qxxi. A Greiger counter contains:

(A) Argon and Alcohol
(B) Alcohol only
(C) Argon only
(D) Water vapours

Qxxii. Three capacitors of 3μF each are connected in series the equivalent capacitance of the circuit is:

(A) 1μF
(B) 3μF
(C) 6μF
(D) 9μF

Qxxiii. If the current passing through a wire held in a uniform magnetic field is doubled, the force acting on the wire will become:

(A) Half
(B) Double
(C) Four times
(D) Six times

Qxxiv. Resistors of 3Ω, 5Ω and 7Ω are connected in parallel. If the P.D across 5Ω resistor is 6V, the P.D across the 7Ω resistors is:

(A) 3V
(B) 6V
(C) 7V
(D) 8V

Qxxv. The radius of the 1st orbit of hydrogen atom is:

(A) 0.53 nm
(B) 0.053nm
(C) 0.0053nm
(D) 0.00053nm

Qxxvi. If the temperature of cold body is decreased, the efficiency of Carnot engine will:

(A) decrease
(B) increase
(C) remain constant
(D) become zero

Qxxvii. Two equal charges each of one coulomb are held 1m apart in free space exert a force of magnitude:

(A) One Newton
(B) 9×10^{9}N
(C) 8.85×10^{-12}N
(D) 6.67×10^{-11}N

Qxxviii. Electrons volt (eV) is a unit of:

(A) electric charge
(B) electric potential
(C) energy
(D) voltage

Qxxix. The smallest total resistance which can be obtained by the combination of 2Ω and 8Ω resistance is:

(A) 1.6Ω
(B) 6Ω
(C) 10Ω
(D) 16Ω

Qxxx. Tesla (T) is the unit of:

(A) potential difference
(B) energy
(C) torque
(D) magnetic flux density

Qxxxi. The core of transformer is laminated to reduce the loss

(A) Flux leakage
(B) Eddy current
(C) copper loss
(D) hysteresis loss

Qxxxii. The mutual induction is the working principle of:

(A) A.C generator
(B) transformer
(C) rectifier
(D) dynamo

Qxxxiii. It is an active device:

(A) resistor
(B) capacitor
(C) inductor
(D) transistor

Qxxxiv. [illegible in source scan]

(A) rotating
(B) accelerating
(C) vibrating
(D) moving with a uniform velocity

Qxxxv. This moves with speed of light:

(A) photon (B) proton

(C) positron (D) electron

Qxxxvi. The minimum energy of a photon required for a pair production is:

(A) 1.02keV (B) 10.2 keV

(C) 102 keV (D) 1020 keV

Qxxxvii. A positron is an anti-particle of:

(A) proton (B) electron

(C) neutron (D) photon

Qxxxviii. LASER is produced due to:

(A) stimulated emission of radiation (B) stimulated absorption of radiation

(C) spontaneous emission of radiation (D) spontaneous absorption of radiation

Qxxxix. In radioactive decay law $N = N_0 e^{-\lambda t}$, λ:

(A) wavelength (B) half-life

(C) decay constant (D) wave number

Qxl. The process in which heavy nucleus splits into two lighter nuclei is called:

(A) fission (B) fusion

(C) mass defect (D) radioactivity

Qxli. The track formed in Wilson Cloud Chamber due to beta particles is:

(A) a thin and broken line (B) at thick and continuous line

(C) scattered dots (D) a thick and broken line

Qxlii. Ammeter is used to measure:

(A) current (B) potential difference

(C) resistance (D) capacitance

Section 'B' (Short-Answer Questions) (25)

Maximum Marks: 25

> Note: Attempt any Five part questions from this section. All questions carry equal marks. The use of scientific calculator is allowed. All notations are used in their usual meanings. Draw diagram where necessary.

Q2.

(i) Show that coefficient of linear expansion is one-third the coefficient of volume expansion.

(ii) How many electrons per second pass through the cross section of a wire carrying a current of 0.7 ampere?

(iii) A galvanometer has a resistance of 50Ω and it deflects full scale when a current of 500 micro ampere flows in it. How can it be converted into ammeter of range 15 ampere and voltmeter of range 300 volts?

(iv) With the help of diagram give the normal working of npn transistor.

(v) Light of wavelength 486.3nm is emitted by a hydrogen atom in Balmer series. What transition of hydrogen atom is responsible for this radiation? ($R_H = 1.097 \times 10^7\ m^{-1}$)

(vi) Sodium surface is shined with light of wavelength 3×10^{-7} m. If the work function of Na is 2.46eV, Find the K.E of the photoelectrons and cut off wavelength λ_c h = 6.63×10^{-34} J.s, c = 3×10^8m/s, 1eV = 1.6×10^{-19} J.

(vii) It is more difficult to start a fusion reaction than the fission reaction why?

(viii) A proton of mass 1.67×10^{-27} kg and charge 1.6×10^{-19} C is to be held motionless between two horizontal parallel plates. The potential difference applied between plates is 1.02×10^{-8} volts. Calculate the distance between plates.

(ix) Give the construction and working of Wilson Cloud chamber.

Section 'C' (Detailed Answer Questions) (18)

Maximum Marks: 18

> Note: Attempt any One question from this section. Draw diagrams, where necessary. The use of scientific calculator is allowed. All notations are used in their usual meanings.

Q3.

(a) What is Carnot engine? Derive an expression for the efficiency of Carnot engine.

(b) What is transformer? On what principle does it work? Derive mathematical relationship between emf and number of turns in the coils.

(c) State the basic postulates of the special theory of relativity. Explain the four important results arrived at by Einstein on the basis of this theory.

Q4.

(a) What is meant by the capacitance of a capacitor? Derive an expression for the the capacitance of a parallel plate capacitor in the presence of air is dielectric between its plates.

(b) What is galvanometer? Prove that current passing through the coil of galvanometer is directly proportional to the angle of deflection of the coil.

(c) State the basic postulates of Bohr's theory of atomic structure. Derive an expression for the radius of the nth orbit of a hydrogen atom.

نوٹ: سال 2020 میں گیارہویں اور بارہویں جماعت کے امتحانات نہیں ہوئے تھے۔